

# System Design (Telusko) - Day 12

One-Shot 5h course | Video 2:45-3:00

## Rate Limiting

Protect services from abuse and overload.

### Key concepts

- Limit requests per client/time window.
- Algorithms: token bucket, leaky bucket, fixed/sliding window.
- Return HTTP 429 when over the limit.
- Store counters in Redis for distributed limits.

### Key example

Token bucket: refill N tokens/sec; each request spends one; empty -> reject.

### Today's tasks (1.5h)

- ☐ Learn token bucket
- ☐ Design an API rate limit
- ☐ Note where to store counters

### Resources

Video: [youtu.be/Vnm-ycSfJx4](https://youtu.be/Vnm-ycSfJx4) (start at 2:45-3:00)

Docs: [docs.telusko.com/docs/system-design](https://docs.telusko.com/docs/system-design)

(Full clickable links are in the app: Goals > this day > Study block.)